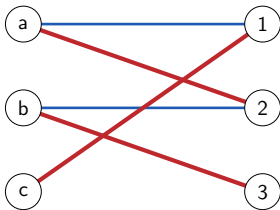


Matchings

Section 1.7

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Definitions

2. Hall's Theorem and SDRs

3. The König–Egerváry Theorem

4. Perfect Matchings

The matching problem

Question

A collection of jobs must be assigned to workers. Each worker can do certain jobs. When can every job be assigned to a different qualified worker?

Represent jobs and workers by the two parts of a bipartite graph. An edge records an allowed assignment.

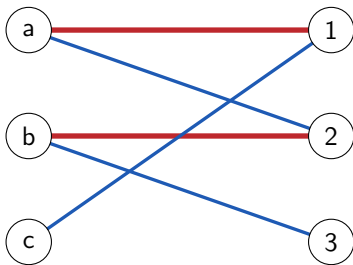
Matchings

Matching

A **matching** in a graph is a set of edges no two of which have a common endpoint.

A vertex incident with an edge of a matching M is **saturated** by M . Otherwise it is **unsaturated**.

An example



The red edges form a matching of size 2.

Maximal and maximum matchings

Two different conditions

A matching is **maximal** if no edge can be added to it. It is **maximum** if no matching has more edges.

In the path $v_1 v_2 v_3 v_4$, the matching $\{v_2 v_3\}$ is maximal. The matching $\{v_1 v_2, v_3 v_4\}$ is maximum.

Perfect matchings

Perfect matching

A matching that saturates every vertex is a **perfect matching**.

A perfect matching in a graph of order n has $n/2$ edges. Thus n must be even.

Even order alone is not sufficient: the star $K_{1,3}$ has no perfect matching.

Alternating and augmenting paths

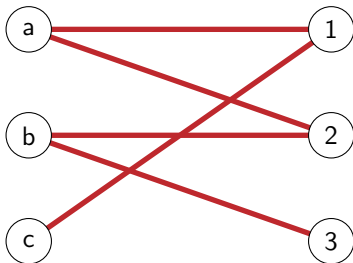
Alternating path

Relative to a matching M , an alternating path has edges alternately outside and inside M .

Augmenting path

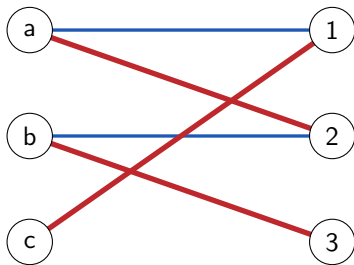
An alternating path whose two endpoints are unsaturated is an **M -augmenting path**. Its first and last edges are outside M .

Finding an augmenting path



$c, 1, a, 2, b, 3$ alternates outside and inside $M = \{a1, b2\}$.

Flipping along the path



Remove $a1, b2$ and add $c1, a2, b3$. The size increases by 1.

Berge's theorem

Berge

A matching M is maximum if and only if there is no M -augmenting path.

Proof

An augmenting path permits a flip that increases the matching size. Therefore a maximum matching has no augmenting path.

For the converse, compare M with a larger matching M' .

Berge's theorem: the converse

Proof

Consider the graph with edge set $M \triangle M'$. Every vertex has degree at most 2, and its nontrivial components are alternating paths or even alternating cycles.

Each cycle has the same number of edges from the two matchings. Since $|M'| > |M|$, some path has more edges from M' than from M .

That path begins and ends with edges of M' , and both endpoints are unsaturated by M . It is an M -augmenting path.

Neighbors of a set

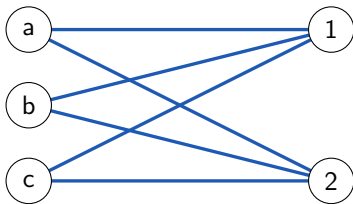
Neighborhood

For a bipartite graph with parts X, Y and a set $S \subseteq X$, let

$$N(S) = \{y \in Y : xy \in E(G) \text{ for some } x \in S\}.$$

To saturate S , its vertices need $|S|$ distinct partners in $N(S)$.

An obstruction



For $S = \{a, b, c\}$, we have $|N(S)| = 2 < 3 = |S|$.

Hall's marriage theorem

Hall

A bipartite graph with parts X, Y has a matching saturating X if and only if

$$|N(S)| \geq |S| \quad \text{for every } S \subseteq X.$$

This condition must hold for every subset, not just for X itself or for individual vertices.

Hall's theorem: necessity

Proof

Suppose a matching saturates X . For any $S \subseteq X$, the matching pairs the vertices of S with distinct vertices of $N(S)$.

Consequently $|N(S)| \geq |S|$.

Hall's theorem: alternating reachability

Proof

Assume Hall's condition, and take a maximum matching M . Suppose $x_0 \in X$ is unsaturated.

Starting at x_0 , follow edges outside M from X to Y , and matching edges from Y to X . Let $S \subseteq X$ and $T \subseteq Y$ be the vertices reached.

No vertex of T is unsaturated: otherwise the search gives an augmenting path, contradicting Berge's theorem.

Hall's theorem: the contradiction

Proof

Every vertex of T has its matching partner in S . Every vertex of $S \setminus \{x_0\}$ is reached through its matching edge and has its partner in T . Hence

$$|S| = |T| + 1.$$

Every neighbor of S lies in T : nonmatching edges are searched, and a matching neighbor is already reached. Also $T \subseteq N(S)$. Thus

$$|N(S)| = |T| = |S| - 1,$$

contradicting Hall's condition.

Systems of distinct representatives

SDR

Given sets $A_1, \dots, A_n$, a **system of distinct representatives** is a choice $a_i \in A_i$ such that the chosen elements are all distinct.

Make a bipartite graph with one vertex for each set and one for each possible element.
Join a set to its elements.

Hall's theorem for sets

SDR form

The sets $A_1, \dots, A_n$ have an SDR if and only if, for every $I \subseteq \{1, \dots, n\}$,

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|.$$

Example

$A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{1\}$ have the SDR $(2, 3, 1)$.

A common mistake

Question

If the union of n nonempty sets has at least n elements, must the sets have an SDR?

No. Take $A_1 = A_2 = \{1\}$ and $A_3 = \{2, 3, 4\}$.

The entire union is large enough, but the first two sets together have only one possible representative.

Regular bipartite graphs

Corollary

Every r -regular bipartite graph with $r \geq 1$ has a perfect matching.

Proof

Let the parts be X, Y . Counting edges gives $r|X| = r|Y|$, so $|X| = |Y|$. For $S \subseteq X$, all $r|S|$ edges at S end in $N(S)$, which can receive at most $r|N(S)|$ edges. Thus $|S| \leq |N(S)|$. Hall gives a matching saturating X and therefore also Y .

Decomposing a regular bipartite graph

Corollary

The edges of an r -regular bipartite graph can be partitioned into r perfect matchings.

Proof

Remove a perfect matching. The remaining graph is $(r - 1)$ -regular and bipartite. Repeat, stopping when no edges remain.

Equivalently, its edges can be properly colored with r colors.

Vertex cover

A **vertex cover** is a set of vertices meeting every edge. Let $\tau(G)$ be its minimum size, and let $\nu(G)$ be the maximum matching size.

Every cover must meet every edge of a matching, using a different vertex for each edge. Therefore

$$\nu(G) \leq \tau(G).$$

The König–Egerváry theorem

König–Egerváry

For every bipartite graph G ,

$$\nu(G) = \tau(G).$$

The bipartite hypothesis matters. In K_3 , a maximum matching has size 1, while a minimum vertex cover has size 2.

Constructing a minimum cover

Proof

Let M be a maximum matching in a bipartite graph with parts X, Y . Start alternating searches at all unsaturated vertices in X . Let Z be the set of vertices reached.

Define

$$C = (X \setminus Z) \cup (Y \cap Z).$$

We will show that C covers every edge and contains exactly one endpoint of every edge in M .

Why the constructed set is a cover

Proof

Suppose an edge xy is not covered. Then $x \in X \cap Z$ and $y \in Y \setminus Z$.

If $xy \notin M$, the search reaches y from x , a contradiction. If $xy \in M$, a reached matched vertex x is reached through this matching edge, so y is reached again.

Thus no edge is left uncovered.

Why the cover has size $|M|$

Proof

No reached vertex in Y is unsaturated, since it would finish an augmenting path. Every unsaturated vertex in X is a search starting point, so it is excluded from C .

For a matching edge xy , its endpoints are either both reached or both unreached. In the first case only y belongs to C ; in the second only x does.

Therefore $|C| = |M|$, proving $\tau(G) \leq \nu(G)$ and hence equality.

The matrix interpretation

Independent entries

In a 0–1 matrix, the largest number of 1's with no two in the same row or column equals the smallest number of rows and columns covering all the 1's.

Use a bipartite graph whose parts are the rows and columns. An entry equal to 1 gives an edge.

Interchanging 0 and 1 gives the corresponding statement about independent zeros.

Matchings as flows

Add a source s and sink t to a bipartite graph with parts X, Y .

- Direct edges $s \rightarrow x$, $x \rightarrow y$, and $y \rightarrow t$.
- Give each edge capacity 1.
- An integral flow of value k specifies a matching of size k .

Max-flow min-cut

The maximum flow value equals the minimum cut capacity. With integer capacities, there is an integral maximum flow.

This gives another route to bipartite matching algorithms and theorems.

Perfect matchings in general graphs

Question

What replaces Hall's condition when the graph is not bipartite?

If S is deleted, an odd component of $G - S$ cannot be matched entirely within itself. It needs an edge to a vertex of S .

Different odd components need different vertices of S .

Tutte's theorem

Odd components

Write $o(H)$ for the number of connected components of H having odd order.

Tutte

A graph G has a perfect matching if and only if

$$o(G - S) \leq |S| \quad \text{for every } S \subseteq V(G).$$

Tutte's theorem: necessity

Proof

Let M be a perfect matching. Each odd component of $G - S$ has a vertex whose matching partner lies in S . Otherwise all its vertices would be paired internally, which is impossible for an odd number of vertices.

Matching edges from distinct odd components use distinct vertices of S . Thus $o(G - S) \leq |S|$.

An obstruction detected by Tutte

Example

Take three triangles and a new vertex v , joined to one vertex of each triangle.

The graph has 10 vertices. Deleting $S = \{v\}$ leaves three odd components:

$$o(G - S) = 3 > 1 = |S|.$$

Hence the graph has no perfect matching, despite having even order.

Tutte's theorem: sufficiency setup

Proof

Use strong induction on the even order of G . The empty graph is the base case. The condition at $S = \emptyset$ implies that every component of G has even order.

Since $|V(G)|$ is even,

$$o(G - S) \equiv |S| \pmod{2}.$$

Call S *tight* if $o(G - S) = |S|$. We distinguish whether there is a nonempty tight set.

Tutte's theorem: no nonempty tight set

Proof

If no nonempty set is tight, parity gives

$$o(G - S) \leq |S| - 2 \quad (S \neq \emptyset).$$

Choose an edge uv ; one exists because a nonempty graph satisfying the condition has no odd isolated component. Let $H = G - u - v$. For every $T \subseteq V(H)$,

$$o(H - T) = o(G - (T \cup \{u, v\})) \leq |T|.$$

By induction H has a perfect matching. Adding uv completes a perfect matching of G .

Tutte's theorem: a largest tight set

Proof

Otherwise choose a nonempty tight set S of largest possible size. There are exactly $|S|$ odd components of $G - S$.

There are no even components: if C were even and $v \in V(C)$, then $C - v$ would have at least one odd component. Thus

$$o(G - (S \cup \{v\})) \geq |S| + 1.$$

Tutte's condition forces equality, contradicting the maximal size of S .

Tutte's theorem: each component is factor-critical

Proof

Let C be an odd component of $G - S$ and fix $v \in V(C)$. For $T \subseteq V(C) \setminus \{v\}$, the larger set $U = S \cup \{v\} \cup T$ is not tight. Parity gives

$$o(G - U) \leq |U| - 2 = |S| + |T| - 1.$$

The other $|S| - 1$ odd components are unchanged, so

$$o(C - v - T) \leq |T|.$$

Induction gives a perfect matching in $C - v$. This holds for every $v \in V(C)$.

Tutte's theorem: use Hall's theorem

Proof

Form a bipartite graph between S and the odd components of $G - S$, joining s to a component when s has a neighbor there.

For $A \subseteq S$, deleting $S \setminus A$ leaves at least $|S| - |N(A)|$ odd components: those with no neighbor in A . Hence

$$|S| - |N(A)| \leq |S| - |A|.$$

Thus $|N(A)| \geq |A|$. Hall pairs every vertex of S with a different component.

Tutte's theorem: finish the matching

Proof

For each $s \in S$, choose an edge sv into its paired component C . Match the remaining vertices of C using a perfect matching of $C - v$.

There are exactly $|S|$ components, so each is used once. These edges and internal matchings are disjoint and saturate every vertex of G .

A useful sufficient condition

Corollary of Dirac's theorem

A simple graph of order $2n$ with minimum degree at least n has a perfect matching.

Proof

For $n \geq 2$, Dirac's theorem gives a Hamilton cycle. Its alternate edges form a perfect matching. For $n = 1$, the degree condition gives the single required edge.

Question

A bipartite graph has parts X, Y of equal size. Every vertex of X has degree at least r , and every vertex of Y has degree at most r , where $r > 0$. Prove that it has a perfect matching.

Which side of the edge count gives an upper bound? Which gives a lower bound?

Proof

For $S \subseteq X$, at least $r|S|$ edges leave S . All end in $N(S)$, whose vertices receive at most $r|N(S)|$ edges in total. Thus

$$r|S| \leq r|N(S)|.$$

Hall's condition follows. The matching saturates X , and since $|X| = |Y|$, it is perfect.

- Augmenting paths characterize maximum matchings.
- Hall's condition characterizes matchings saturating one part of a bipartite graph.
- In bipartite graphs, maximum matching size equals minimum vertex cover size.
- Tutte's odd-component condition characterizes perfect matchings in general graphs.

Ramsey Theory